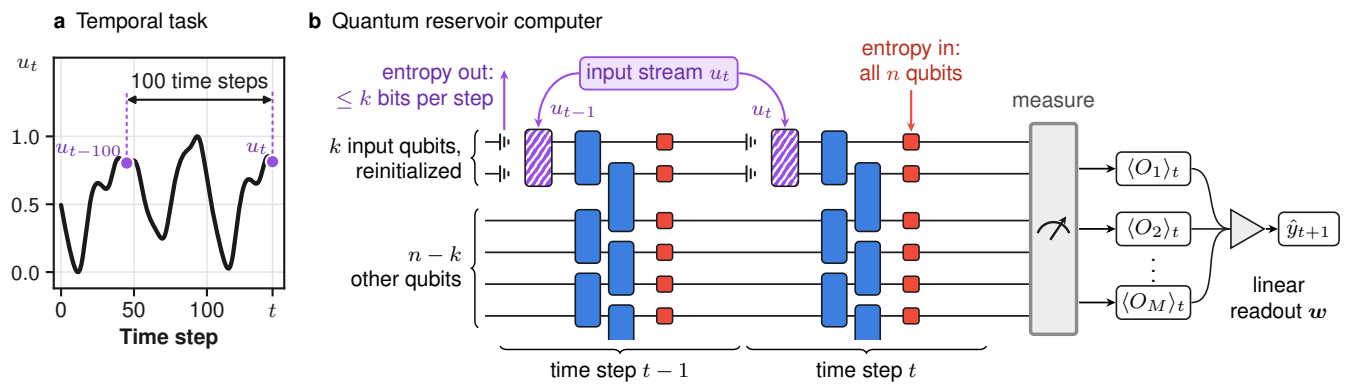

a Temporal task
u_t
100 time steps
1.0
0.5
0.0
u_{t-100}
u_t
0
50
100
t
Time step
b Quantum reservoir computer
entropy out:
$\leq k$ bits per step
input stream u_t
u_{t-1}
u_t
entropy in:
all n qubits
measure
k input qubits,
reinitialized
$n-k$
other qubits
$\langle O_1\rangle_t$
$\langle O_2\rangle_t$
$\langle O_M\rangle_t$
$\hat{y}_{t+1}$
linear
readout $\boldsymbol{w}$
time step $t-1$
time step t
target of the forecast:
$y_{t+1} = 0.7\,u_t + 0.3\,u_{t-100}$
reinitialize in $\sigma(u_t)$,
e.g. $R_y(u_t)|0\rangle$ per qubit
fixed reservoir unitary U,
any structure
noise on every qubit:
$\mathcal{D}_q(\rho) = q\rho + (1-q)\,I/2$